THEXRA ENTERPRISE SOLUTION PROPOSAL

Architecture Recommendation Brief

Date: 02/10/2026
Status: Draft

1. CLIENT PROBLEM STATEMENT

High factory equipment downtime and severe safety risks during heavy machinery maintenance procedures.

2. AI RATIONALE & ANALYSIS

"Implementing a combined AR and predictive platform directly eliminates maintenance downtime and safety hazards by providing technicians hands-free digital checklists and live remote expert guidance on-site. THEXRA Suite delivers the end-to-end industrial software stack required to create, deploy, and execute spatial maintenance procedures securely across heavy factory environments."

3. PROPOSED SOLUTION CONCEPT

An integrated enterprise industrial platform combining hands-free augmented reality guidance, remote expert collaboration, and predictive safety telemetry for heavy machinery maintenance.

IMPLEMENTATION APPROACH:

1. Phase 1: Operational audit, hazard mapping, and network connectivity infrastructure readiness.
2. Phase 2: Authoring digital standard operating procedures and configuring edge-to-cloud remote expert endpoints.
3. Phase 3: Controlled pilot deployment on priority machinery lines followed by site-wide industrial rollout.

DETAILED IMPLEMENTATION PLAN:

1. **Discovery** (3 weeks):Site technical readiness evaluation • Machinery maintenance workflow and hazard map
2. **Design** (4 weeks):Digital SOP workflow diagrams • Spatial user interface wireframes and headset interaction maps
3. **Prototype** (5 weeks):Proof-of-concept maintenance pilot on single machine line • Field user feedback and technical viability report
4. **Development** (10 weeks):Complete THEXRA Suite digital procedure library • Remote expert audio-visual connectivity integration • Enterprise IoT telemetry connectors
5. **Testing** (5 weeks):End-to-end hardware stress and safety compliance testing • User acceptance testing sign-off from maintenance leads
6. **Deployment** (5 weeks):Production deployment across targeted factory plant zones • Hardware provisioning and technician certification training
7. **Support** (Ongoing):24/7 technical system support and maintenance SLA • Monthly machinery downtime reduction and utilization reports

SCOPE OF WORK (SOW)

1. PROJECT SCOPE

Delivery of an integrated enterprise maintenance system combining AR headsets, THEXRA Suite applications, and remote expert assistance for heavy machinery facilities.

2. CORE FEATURES

- Hands-free AR step-by-step maintenance checklists
- Live remote expert video calling with 3D spatial annotations
- Automated safety compliance verification and audit logging
- Digital workflow authoring via THEXRA Make

3. DELIVERABLES

- Configured THEXRA Suite instance (Remote, Make, View)
- Digitized standard operating procedure library for key machinery
- Enterprise management dashboard and reporting system

- Hardware setup and provisioning package

4. HARDWARE REQUIREMENTS

- Industrial ruggedized AR smart glasses
- On-premise edge computing gateways
- Dedicated charging and sanitization docking stations

5. SOFTWARE MODULES

- THEXRA Suite enterprise licenses
- Machinery telemetry gateway integration software
- Central web administrative console

6. CONTENT ASSETS

- Interactive 3D spatial guides for heavy equipment
- High-resolution digital safety and maintenance documentation
- Admin user guides and step-by-step authoring manuals

7. TRAINING PLAN

Role-specific training including hands-on operational workshops for floor technicians, expert console training for senior engineers, and CMS authoring sessions for safety managers.

8. DEPLOYMENT STRATEGY

Phased rollout starting with a validated single-line pilot before scaling across all designated heavy manufacturing plant cells.

9. ONGOING SUPPORT

Tier-2 and Tier-3 technical support, scheduled monthly software optimizations, quarterly feature releases, and defined 4-hour SLA for critical incidents.

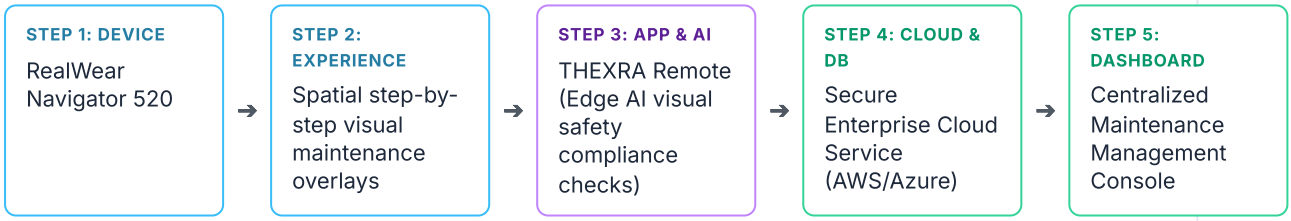
10. EXCLUSIONS (OUT OF SCOPE)

- Physical repair or mechanical overhaul of factory machinery
- Plant-wide structural electrical or facility Wi-Fi re-cabling
- Third-party ERP/MES custom schema overhauls outside standard APIs

4. TECHNOLOGY ARCHITECTURE BLUEPRINT

Combination solution — THEXRA Suite (Remote, Make, View)

AUTOMATED SYSTEM FLOW DIAGRAM:



1. Overview: A hybrid industrial architecture deploying AR smart glasses and Edge AI processing to deliver real-time visual instructions and remote assistance directly to shop floor personnel. The system synchronizes machine telemetry and digital standard operating procedures via a central cloud backend to ensure compliance and machine uptime.

2. User Journey: Technician wears AR headset → Selects heavy machinery maintenance routine via voice or gaze → Receives 3D spatial step-by-step safety prompts → Calls remote specialist when an anomaly occurs → Specialist annotates live field view → Task is logged and digitally signed off.

3. Hardware: RealWear Navigator 520, HTC VIVE Eagle, On-premise Edge AI Gateways

4. Software: THEXRA Remote, THEXRA Make, THEXRA View

5. AI Components: Edge AI visual safety compliance checks, Predictive machinery failure pattern detection

6. XR Components: Spatial step-by-step visual maintenance overlays, Real-time bidirectional telestration and AR call annotation

7. Backend: Secure Enterprise Cloud Service (AWS/Azure), Industrial IoT MQTT Broker and REST APIs

8. Dashboard: Centralized Maintenance Management Console, Real-Time Field Safety and Telemetry Dashboard

9. Data Flow: Smart Glasses / Sensors → Edge AI Gateway → Secure Cloud Backend → Remote Expert Dashboard & Headset Display

Alternative Option: AR — THEXRA Remote

5. EXPECTED BUSINESS OUTCOMES

Up to 40% reduction in equipment downtime, 50% faster issue resolution, and a significant decrease in hazardous site incidents through hands-free SOP execution.

6. IMPLEMENTATION NEXT STEPS

Conduct an on-site facility assessment and safety workflow audit to prioritize critical machine lines for AR digital procedure capture.